Quadratic-twist registry: cover catalog refresh

2026-10-22 — bounded correctness and metadata consistency

The native quadratic-twist study added 1,921 exact models and 27 selected EC1 representations. This follow-up updates the existing native cover catalog and its assessment for all 4,493 registry models. It corrects CI integration failures; it supplies no new transfer algorithm or solver claim. The input registry is unchanged from commit `4711f1c73887`.

Profile	Models	Same-field construction
Ordinary binary	165	genus 3, degree 3
Direct odd prime	2,011	genus 2, degree 2
Normalized odd prime	30	genus 2, degree 2; explicit normalization
Direct odd extension	2,287	genus 2, degree 2 over the declared extension
Total	4,493	4,491 distinct cover models

The exact graph binds 538 declared fields, 4,491 Jacobian contexts, 4,493 maps, and 1,004 selected EC1 representations (140 binary and 864 odd-characteristic). Missing generators remain unknown. One assessment profile covers every map and every EC1/curve/map triple; classification is not blanket arithmetic support.

Evidence. The native Rust checker recomputes equations, smoothness and field checks within its stated boundary, and exact HC1/CV1/ICV1/EC1 links. The standalone test retains every per-model assertion and now visits all 4,493 models. The separate assessment verifier checks every formula, exact membership, typed context, and source digest. Commands, failed reproductions, source hashes, and replay outputs are retained in the adjacent validation record. The complete 434-field native twist replay already passed CI, with the independent IC audit, corruption controls, metadata replay, and receipt upload.

Boundaries. Odd base-prime validity remains the checker's explicit primality-screen assumption. Cover geometry is not an induced Jacobian map, exact subgroup-order proof, or efficient DLP transfer. Catalog-wide arbitrary norms, independent all-element map oracles, odd balanced-Jacobian maps, and proper-subfield descent remain open. The 27 added EC1 records do not promote the full odd profile to arithmetic or transfer support.

The exact fixed-GF(8) arbitrary-class fixture remains outside this catalog. The old packaged artifact was replayed on 592 bundled rows; its historical 746/1,354 allowlist comparison is not an expanded-catalog artifact replay. No performance, general GHS, or general ECDLP hardness conclusion follows. The twist-to-Frobenius-negation proof and shared Conductor access remain open.

Provenance and reproduction. The prior dated assessment report, its graphs and PDF, frozen twist results, and historical receipts remain intact. Run `cargo run -release -locked -bin curve_cover_check - -check`, the standalone constructor tests, the deterministic assessment check, and its independent verifier. The source-linked README and validation record give exact commands. Canonical IC/rho cost graphs are unchanged: no cost comparison was run.

Map and subgroup boundaries

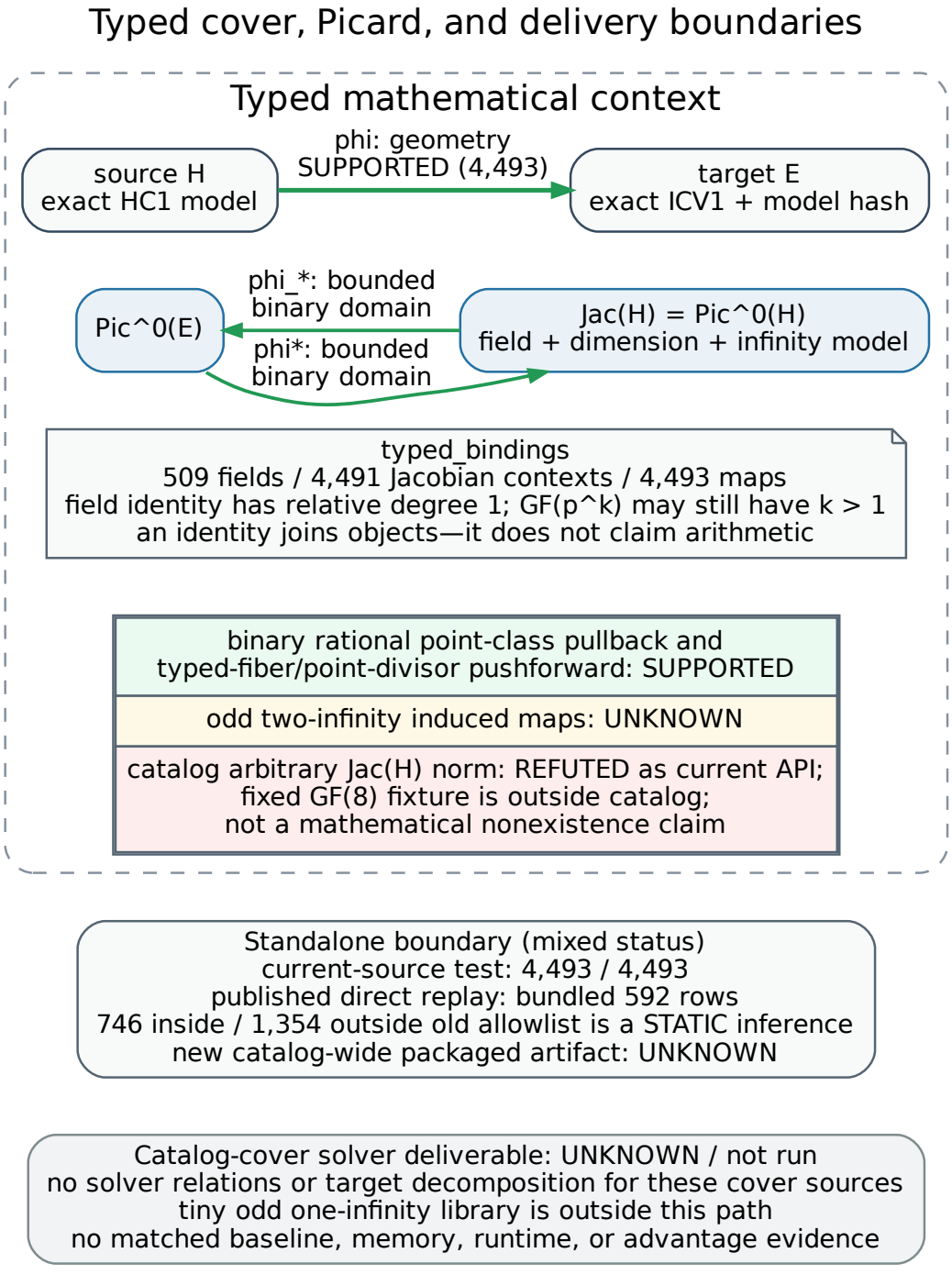
Catalog-wide profile assessment (2026-10-22)

Exact current catalog
2,572 map rows / 4,491 cover models / 1004 EC1 representations
all rows classified; classification is not blanket support

Map assessments: 4,493 / 4,493 exact rows		
Profile	**SUPPORTED**	**Explicit boundary**
165 binary g3, degree 3	construction + geometry one-infinity Jacobian typed point-class pull/push	all-element oracle: UNKNOWN catalog arbitrary Jac(H) norm: REFUTED as current API fixed GF(8) fixture is outside catalog
2,011 odd prime direct	construction + geometry	two-infinity maps: UNKNOWN one-infinity use: REFUTED
30 odd prime normalized	construction + geometry degree-one normalization	coordinate commutation + balanced maps: UNKNOWN
2,287 GF(p^k)	same-declared-field construction + geometry	proper-subfield descent + balanced maps: UNKNOWN

Subgroup assessments: 1004 / 1004 exact EC1 / curve / map triples		
Profile	**SUPPORTED**	**UNKNOWN unless overlaid**
140 binary	identity, gcd(3,r), G on E, [r]G=O pullback, annihilation, composition, degree-inverse recovery of tested G	exact ord(G)=r; additivity; all-element injectivity; image order; arbitrary-result recovery
exact binary overlay	icv1-f2m7-t13-616700dd generator + image order 29	only that EC1/map triple
864 odd	identity + gcd(2,r)	generator/order evidence; every induced-map claim
exact odd overlay	icv1-fp10-t5-192cb216 map geometry + order-823 identity	no subgroup preservation follows

SUPPORTED on labeled domain	UNKNOWN / conditional	REFUTED implementation claim	MIXED: inline labels govern


Typed cover, Picard, and delivery boundaries
Typed mathematical context
source H
exact HC1 model
phi: geometry
SUPPORTED (4,493)
target E
exact ICV1 + model hash
Pic^0(E)
phi_*: bounded
binary domain
phi*: bounded
binary domain
Jac(H) = Pic^0(H)
field + dimension + infinity model
typed_bindings
509 fields / 4,491 Jacobian contexts / 4,493 maps
field identity has relative degree 1; GF(p^k) may still have k > 1
an identity joins objects—it does not claim arithmetic
binary rational point-class pullback and
typed-fiber/point-divisor pushforward: SUPPORTED
odd two-infinity induced maps: UNKNOWN
catalog arbitrary Jac(H) norm: REFUTED as current API;
fixed GF(8) fixture is outside catalog;
not a mathematical nonexistence claim
Standalone boundary (mixed status)
current-source test: 4,493 / 4,493
published direct replay: bundled 592 rows
746 inside / 1,354 outside old allowlist is a STATIC inference
new catalog-wide packaged artifact: UNKNOWN
Catalog-cover solver deliverable: UNKNOWN / not run
no solver relations or target decomposition for these cover sources
tiny odd one-infinity library is outside this path
no matched baseline, memory, runtime, or advantage evidence